

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I Priyanga.B(192511102)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel

values: $r=[50,100,150,200]$

Compute the transformed values.

(a) Purpose

Grey level transformation modifies pixel intensity values to enhance image quality improving contrast, brightness, or highlighting specific intensity ranges without changing spatial relationships between pixels. It is a point processing technique where each output pixel depends only on the corresponding input pixel.

(b) Negative Transformation

Formula: $s = 255 - r$

r	$255 - r$	s
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Answer: 205,155,105,55

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s=c\log(1+r)$, where $c=10$, compute

s for: $r=[0,10,50,100]$

(a) Purpose

Log transformation compresses the dynamic range of pixel values, expanding dark (low-intensity) pixel values while compressing bright (high-intensity) ones. Since the log curve rises steeply at low r and flattens at high r , it enhances details in dark regions of the image (e.g., in Fourier spectrum display or underexposed images).

(b) Calculation

Formula: $s = c \cdot \log(1 + r)$, $c = 10$ (natural logarithm)

r	$1 + r$	$s = 10 \times \ln(1 + r)$
0	1	0.00
10	11	23.98
50	51	39.32
100	101	46.15

Answer: 0.00, 23.98, 39.32, 46.15

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

(a) Purpose

Histogram equalization redistributes pixel intensities so the output histogram is approximately uniform, spreading out the most frequent intensity values. This increases global contrast, especially useful when an image's usable data is represented by close contrast values.

(b) Equalization

Given Histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Total pixels = 16

Step 1: Cumulative Distribution

Intensity	Frequency	Cumulative
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Intensity — using $(L - 1)/N \times \text{CDF}$, with $L = 8$, $N = 16$:

Intensity	CDF	Equalized Value
0	2	$0.875 \approx 1$
1	4	$1.75 \approx 2$
2	8	$3.5 \approx 4$
3	16	$7 \approx 7$

Final Answer

Original	Equalized
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region: 10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

(a) Purpose

Spatial smoothing filters reduce noise and blur fine details by averaging pixel values within a neighborhood, replacing sharp intensity variations with a smoother transition commonly used for noise reduction and pre-processing before edge detection.

(b) Average Filter

Image Matrix:

10	20	30
20	30	40
30	40	50

$$\text{Sum} = 10+20+30+20+30+40+30+40+50 = 270$$

$$\text{Average} = 270 / 9 = 30$$

Answer: New center pixel = 30

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$, $\begin{bmatrix} -1 & 4 & -1 \end{bmatrix}$, $\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

(a) Purpose

Sharpening filters such as the Laplacian highlight regions of rapid intensity change (edges) by computing the second derivative of the image. Adding this result to the original image emphasizes fine detail and edges that smoothing removes.

(b) Laplacian Mask

Mask:

0	-1	0
-1	4	-1
0	-1	0

Image:

10	10	10
10	20	10
10	10	10

$$\text{Calculation: Output at center} = 4(20) - (10+10+10+10) = 80 - 40 = 40$$

Answer: Output at the center = 40